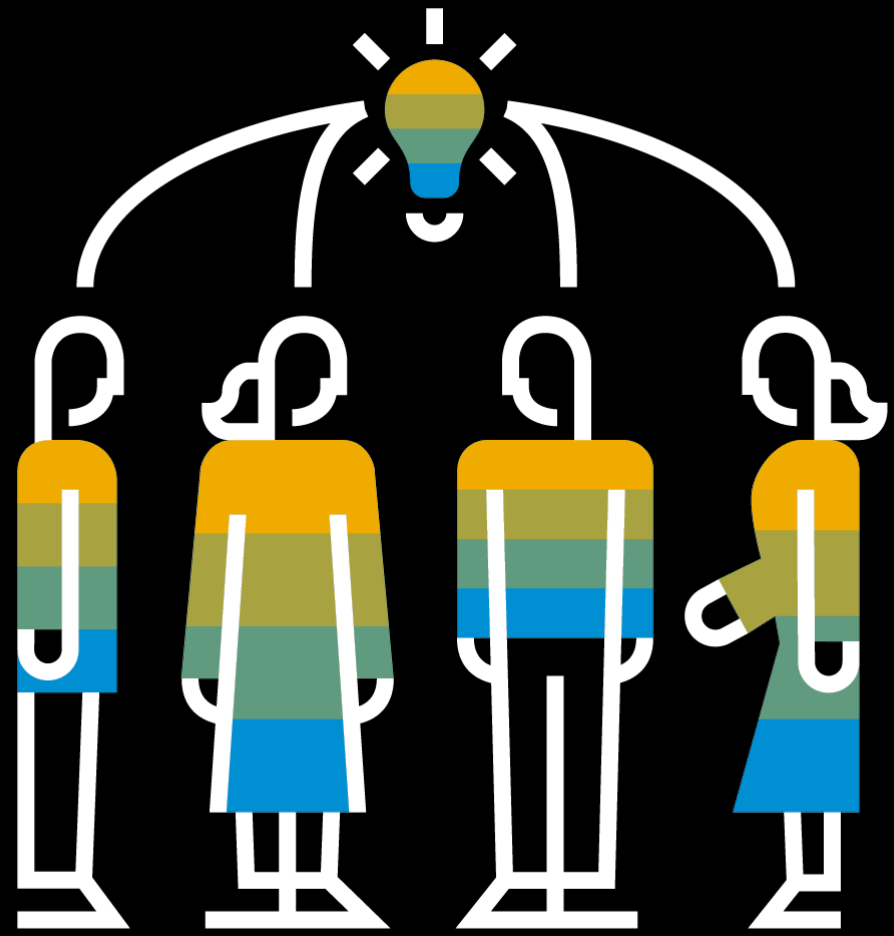
BRIM S/4HANA

Subscription Order Management – Device as a Service – Equipment Integration

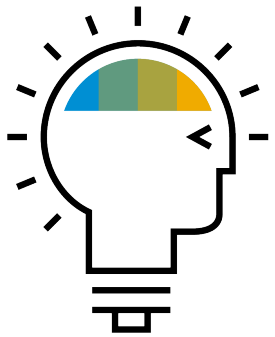
SAP S/4HANA for billing and revenue innovation management
Subscription Order Management
in S/4HANA Service (S/4HANA 2020 FSP00)

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Objectives



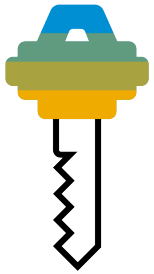
- Get to know the concept of linking equipment to subscription contracts
- Learn how equipment can influence the contract activation and the related change processes

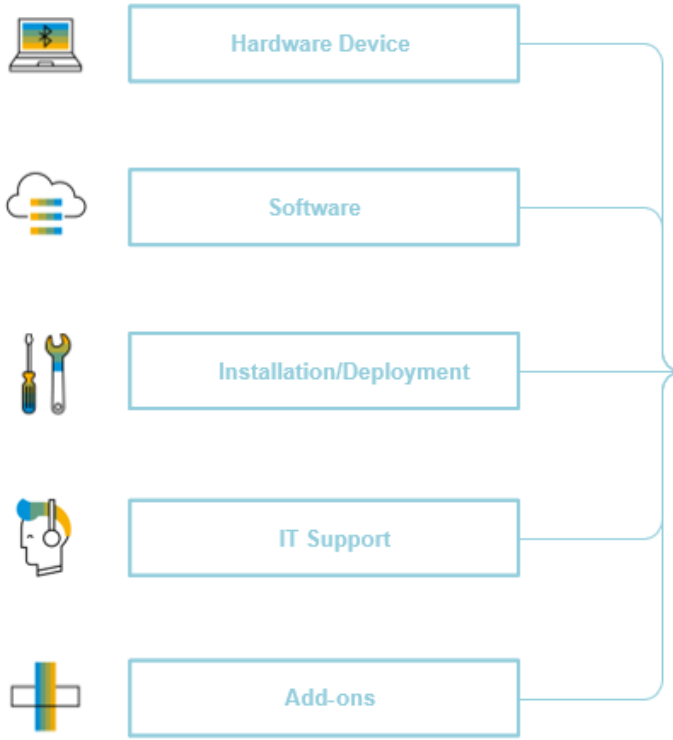
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Key Facts Device as a Service (DaaS)

Definition

Service Providers take a **typical hardware device (such as a laptop, desktop, tablet, printer or mobile phone), bundle it with a variety of services and software**, **and offer it to a customer for a monthly subscription fee**. It gives customers one contract and just one provider to hold accountable.

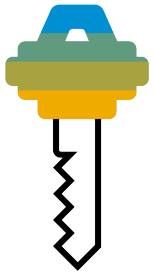




Solution Quotations already combine these services. In addition equipment masters can be directly linked to subscription items.

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Features Device as a Service (DaaS)



Features

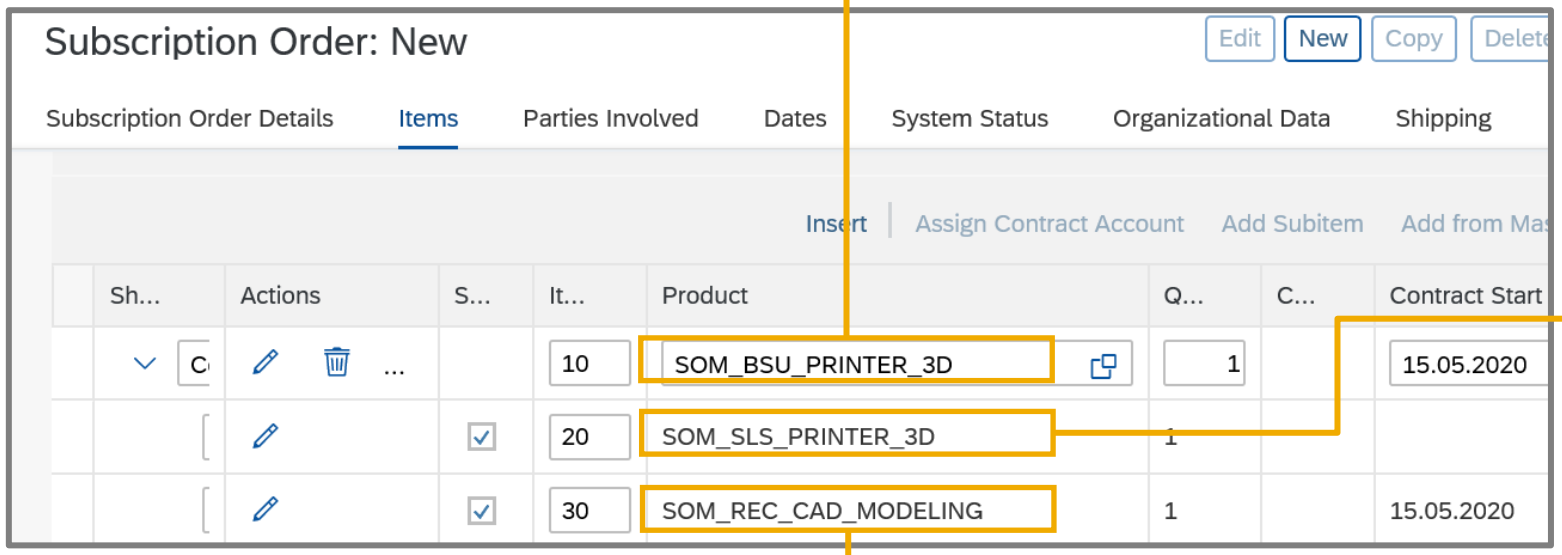
- Linking of equipment to subscription contracts
- Early Distribution
- Distribution and Activation based on equipment activation
- Billing Start Date based on equipment activation
- Equipment ID as Technical Resource
- Change Processes – Replace and Scale

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Device as a Service:
Equipments in Subscription Items



Sample Product



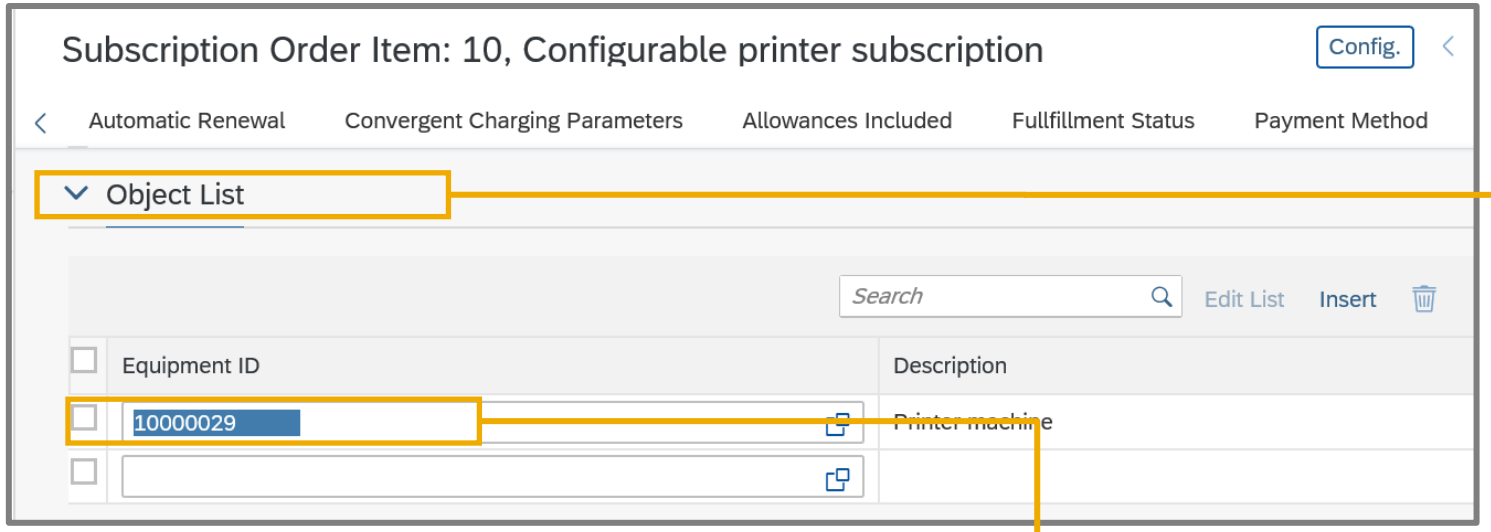
Product that represents the Device Subscription that results in a subscription contract. The equipment ID(s) are to be maintained against this item.

Product that represents the Device Hardware that results in a Sales Order. The relevant equipment are determined based on the Serial Numbers that are assigned in Delivery. Thus, determined equipment ID(s) are to be maintained on the related Subscription Item.

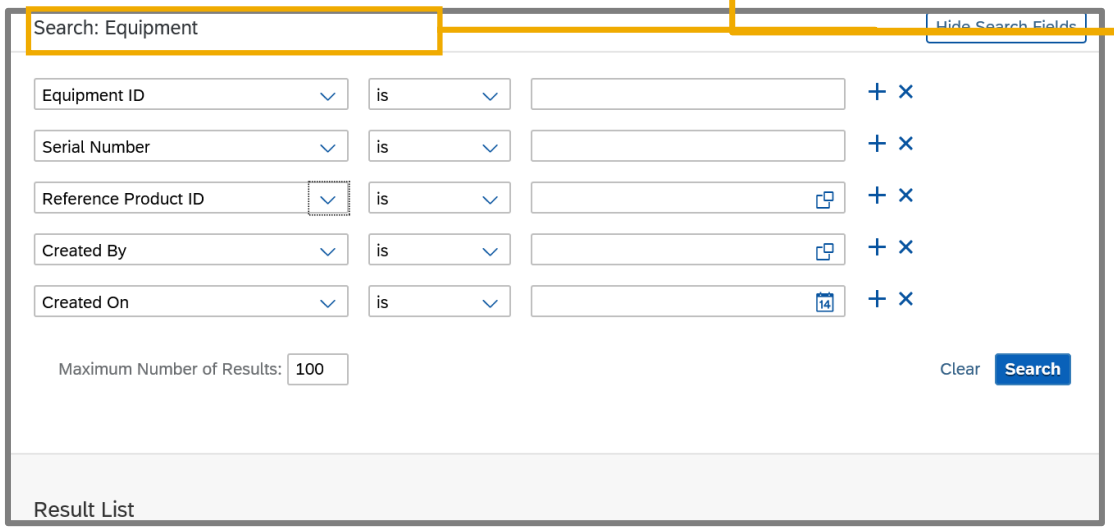
Product that represents a related Subscription, that is dependent on the Device, results in a subscription contract.

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Equipment IDs on Subscription Item



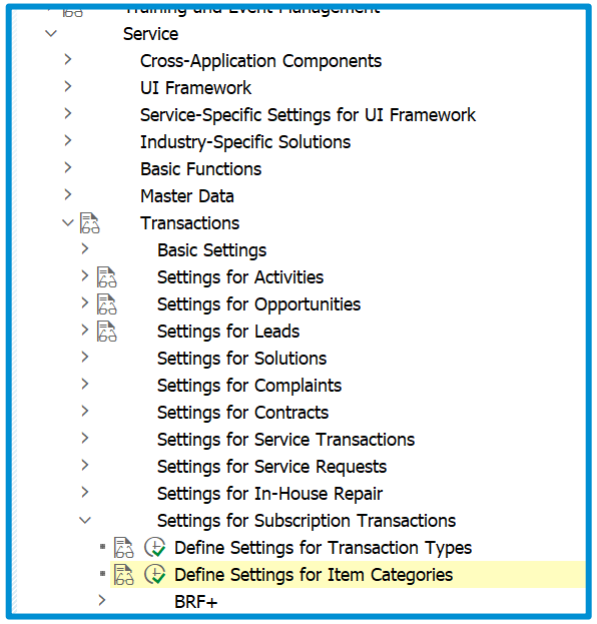
New assignment block is available on the subscription item for maintenance of equipment ID(s).



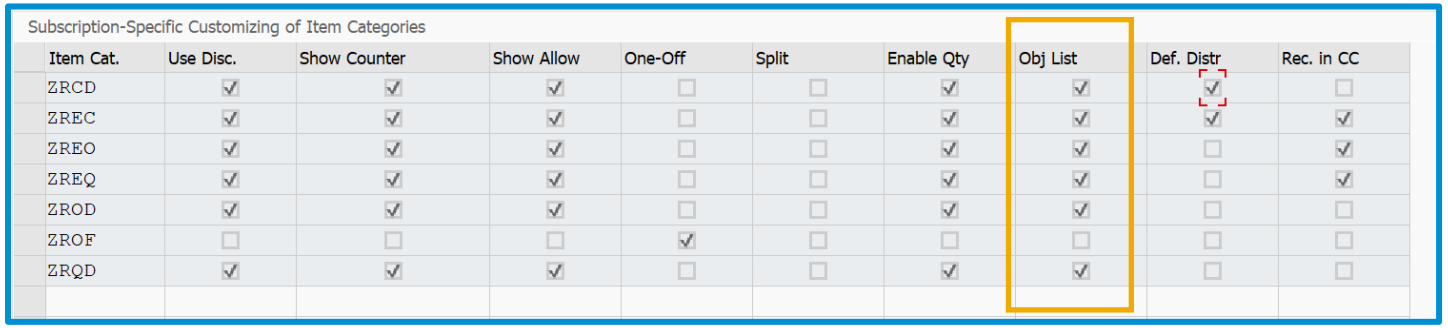
Search based on serial number, reference product etc. is possible.

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Customizing for Object List in Subscription Items



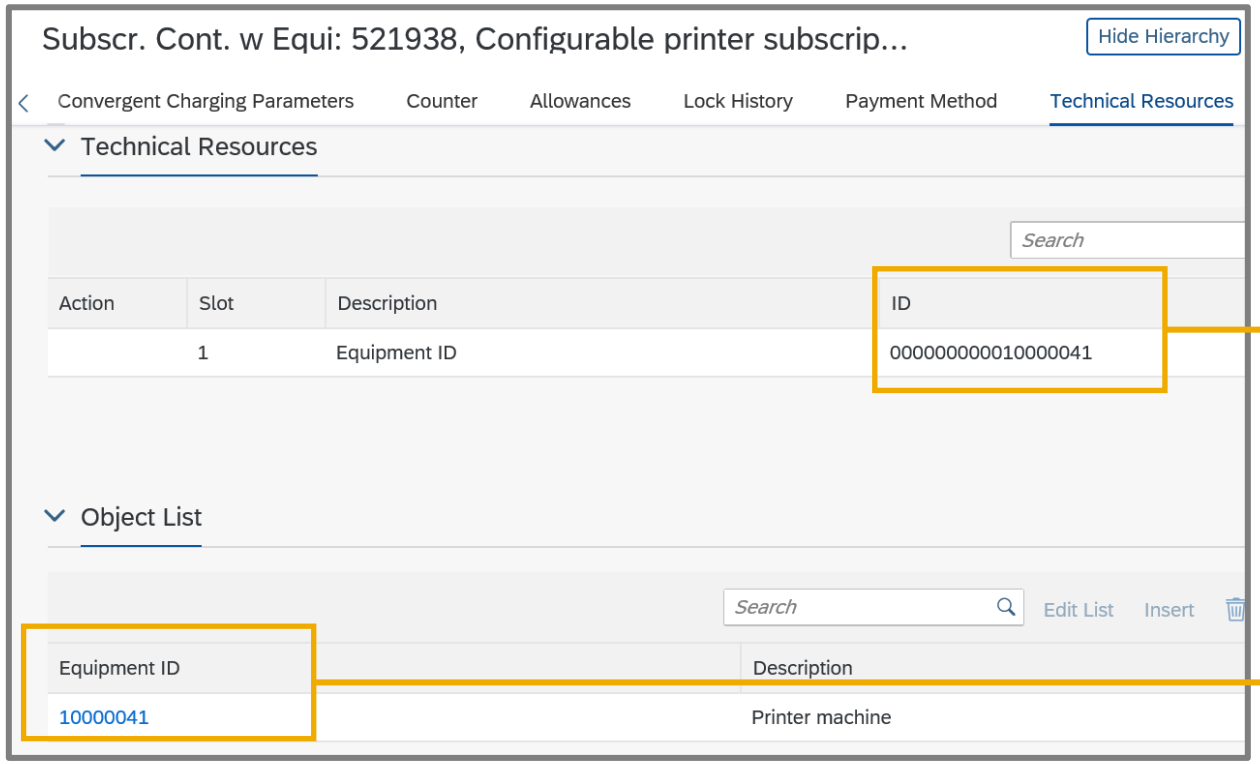
Service -> Transactions -> Settings for Subscription Transactions -> Define Settings for Subscription Items -> Define Settings for Item Categories

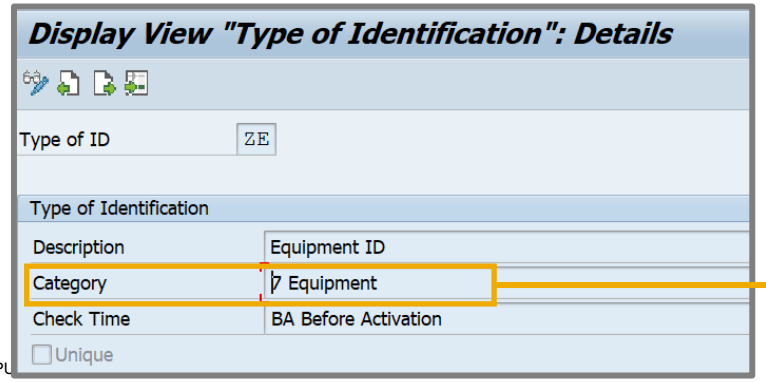


Equipment can be assigned to subscription items in assignment block object List.

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Equipment ID as Technical Resource





It might be required for distribution of equipment IDs to the CC contract, where it can be used in the calculation of usage-based charges.

To facilitate this, equipment IDs maintained in the object list are copied to technical resources and distributed to the CC contract, if a resource type of category equipment is configured for the product.

The technical resource type should be defined with the category Equipment.

SPRO -> Service Transactions -> Settings for Subscription Transactions -> Technical Resources -> Define Types of Identification

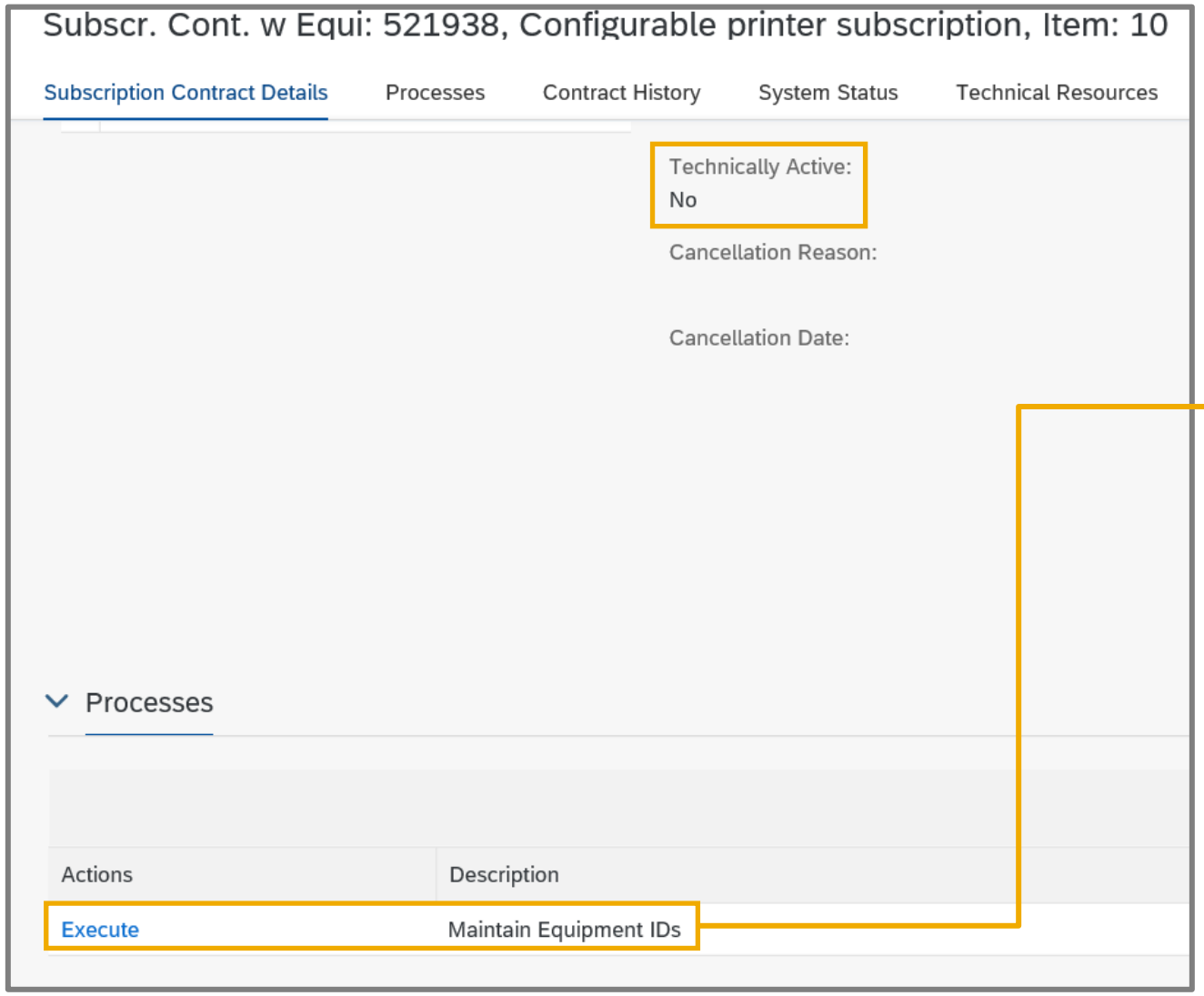
Note: if for example serial numbers are different from equipment IDs and are to be used as technical resources instead of equipment IDs then a BADI is available

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Device as a Service:
Assign equipment and distribute Contract



Equipment IDs on inactive Subscription Contract



Typically, the equipment ID(s) might not be known at the time of creation of the subscription order / solution quotation. So the subscription contract will be created without equipment ID(s) but remain inactive.

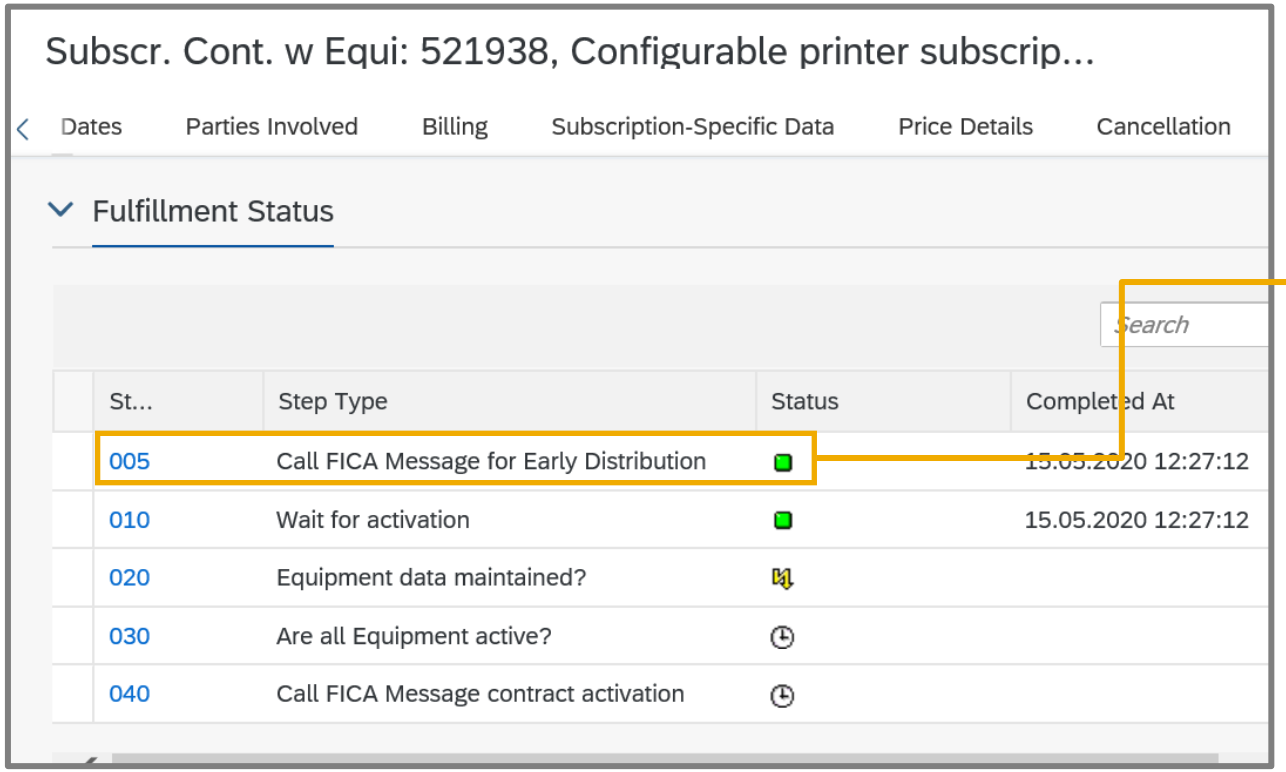
Equipment ID(s) that are determined after downstream processes, such as delivery of the hardware, need to be maintained on the subscription contract.

To support such late maintenance of equipment ID(s), a new change process ‘Maintain Equipment IDs’ is available.

This process can be used as long as the contract is not active, preceding document is not open (e.g. solution quotation) and distribution is not currently in progress.

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Early Distribution



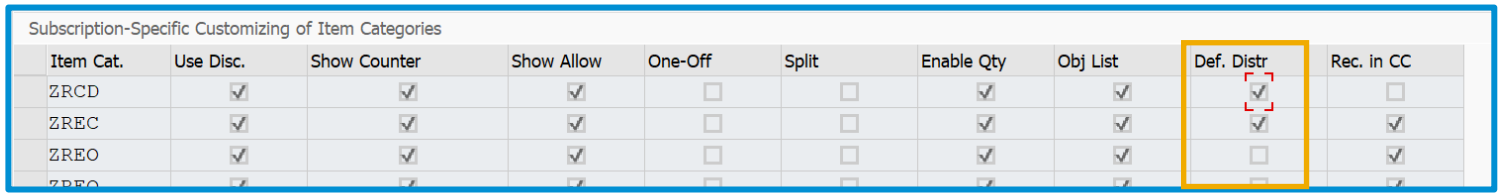
Typically, early distribution of contract comes into picture only when the start date is in the future.

Now we have the scenario of where the contract start date might be immediate but the actual activation and distribution should happen only once the related equipment is activated.

To support this, a feature has been made available that allows for the execution of early distribution even when the start date is immediate.

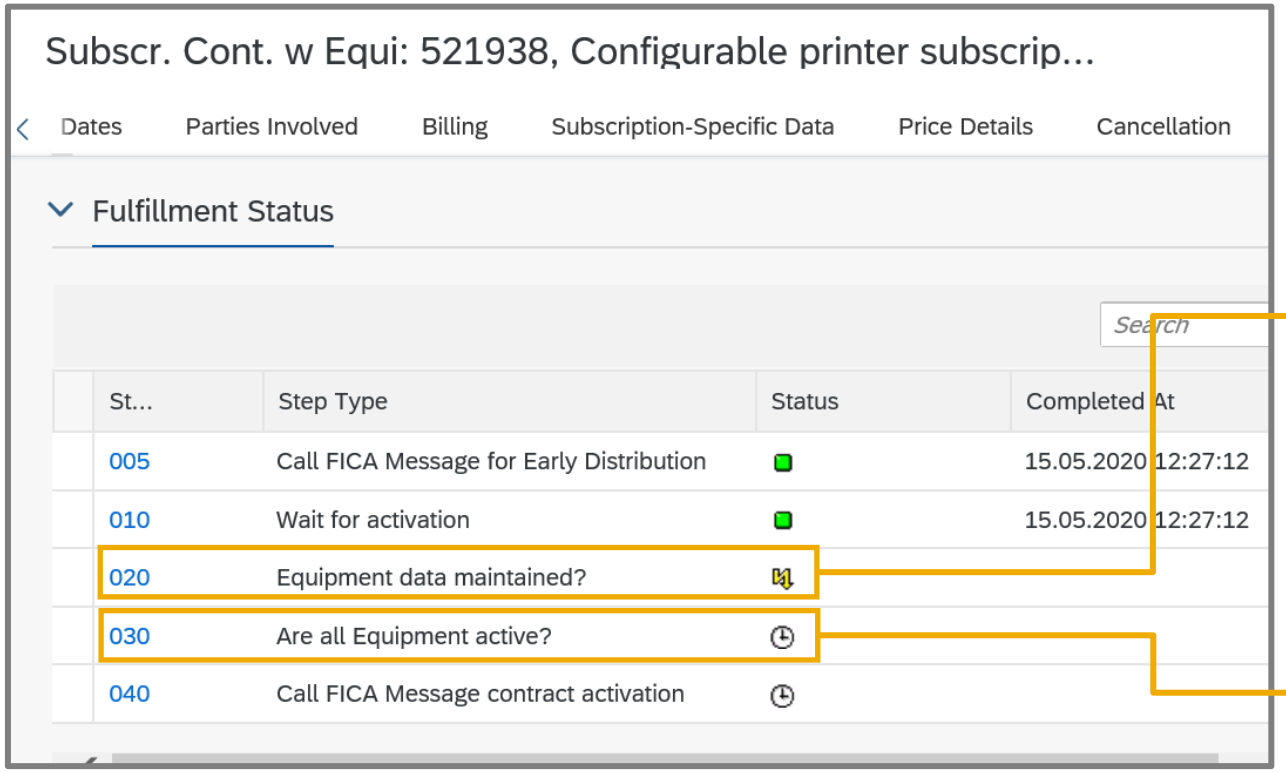
This feature can be configured at item category level Service -> Transactions -> Settings for Subscription Transactions -> Define Settings for Subscription Items -> Define Settings for Item Categories

Flag Deferred Distribution



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Distribution and activation based on Equipment activation



Subscription contract will remain inactive until the related hardware has been delivered and activated.

This is achieved by using the ODI framework.

Equipment ID(s) may be determined based on downstream processes such as delivery. Until such determined equipment ID(s) are maintained on the contract, the step will not be successfully executed.

This step will not be successfully executed until a particular status (e.g. At Customer) is set on the equipment.

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Customizing ODI for equipment

Schema Steps

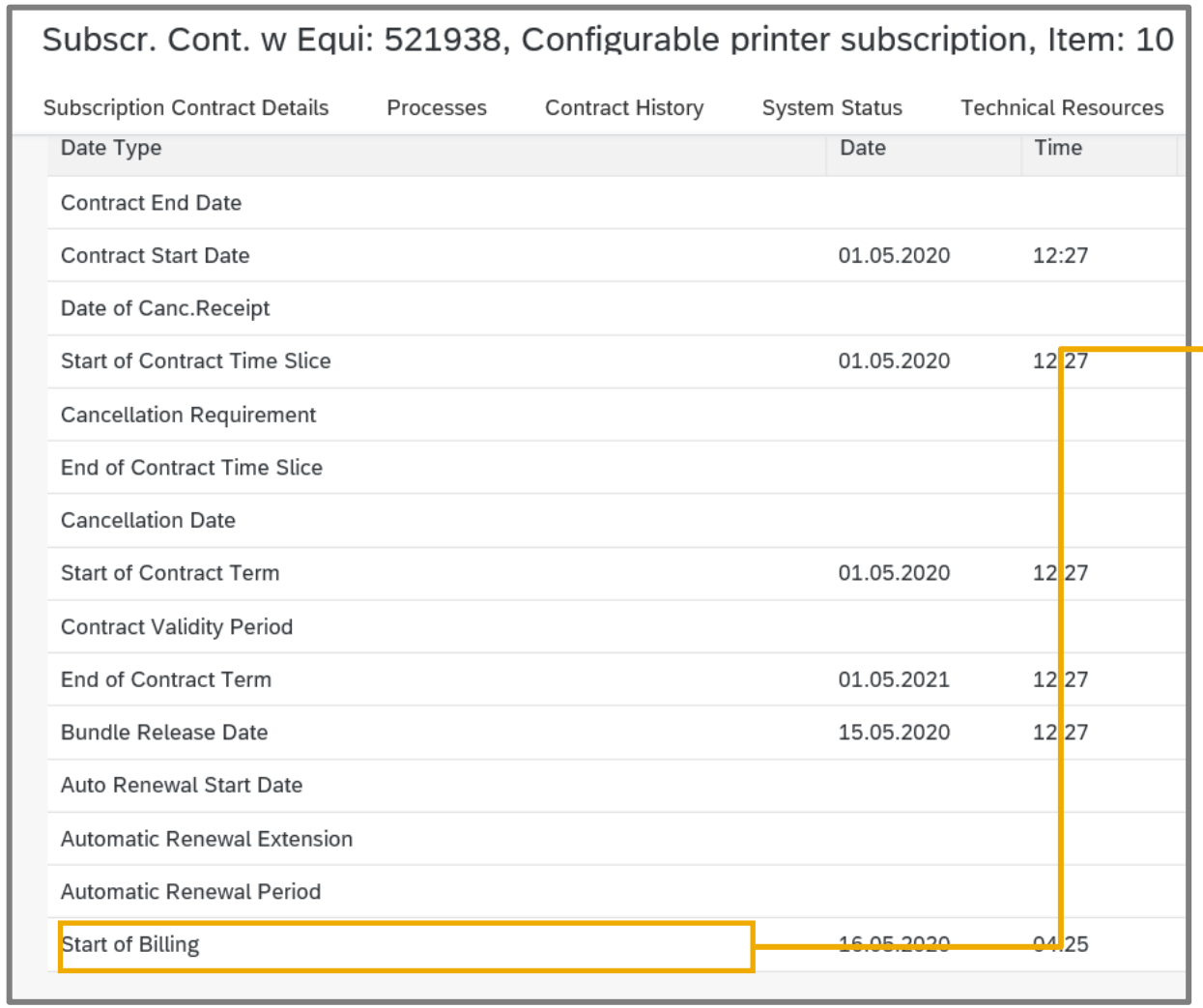
Step ID	Step Description	Step Type	Document Distribution Step Ty...	Category	Distribution Step Category	Step Seq.	PrereqStep	Inactive	Batch
EQUA	Are all Equipment active?	EQUA	Are all Equipment active?	P1	Subscription Contract Activation	30	20	☐	☐
EQUM	Equipment data maintained?	EQUM	Equipment data maintained?	P1	Subscription Contract Activation	20		☐	☐
EQUW	Wait for Contract Start date	EQUW	Wait for Contract Start date - no...	P0	Subscription Contract waits for a...	20	10	☐	☐
PCEA	Call FICA Message contract activ...	PCEA	Call FICA Message contract activ...	P1	Subscription Contract Activation	40	30	☐	☐
PCER	Call FICA Message for Early Distr...	PCER	Call FICA Message for Early Distr...	P0	Subscription Contract waits for a...	10		☐	☐

There might also be business scenario when the equipment ID is not known upfront. The hardware device would be delivered to the customer and the activation of subscription contract would be dependent on the equipment delivery status. Example implementation has been provided considering this scenario where the physical device is a sales sub-item of the contract main item. The sales item is processed for delivery where the equipment ID, linked to the serial number of the sales product, is assigned in delivery document. On POST GOODS ISSUE of the delivery document the equipment status is updated to ECUS.

ODI schema EQUI has been delivered with the following steps which get updated based on certain events when delivery document is created and processed.

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Billing Start Date based on Equipment activation



Typically, the date from which the recurring fees are applicable is the same as the Contract Start Date.

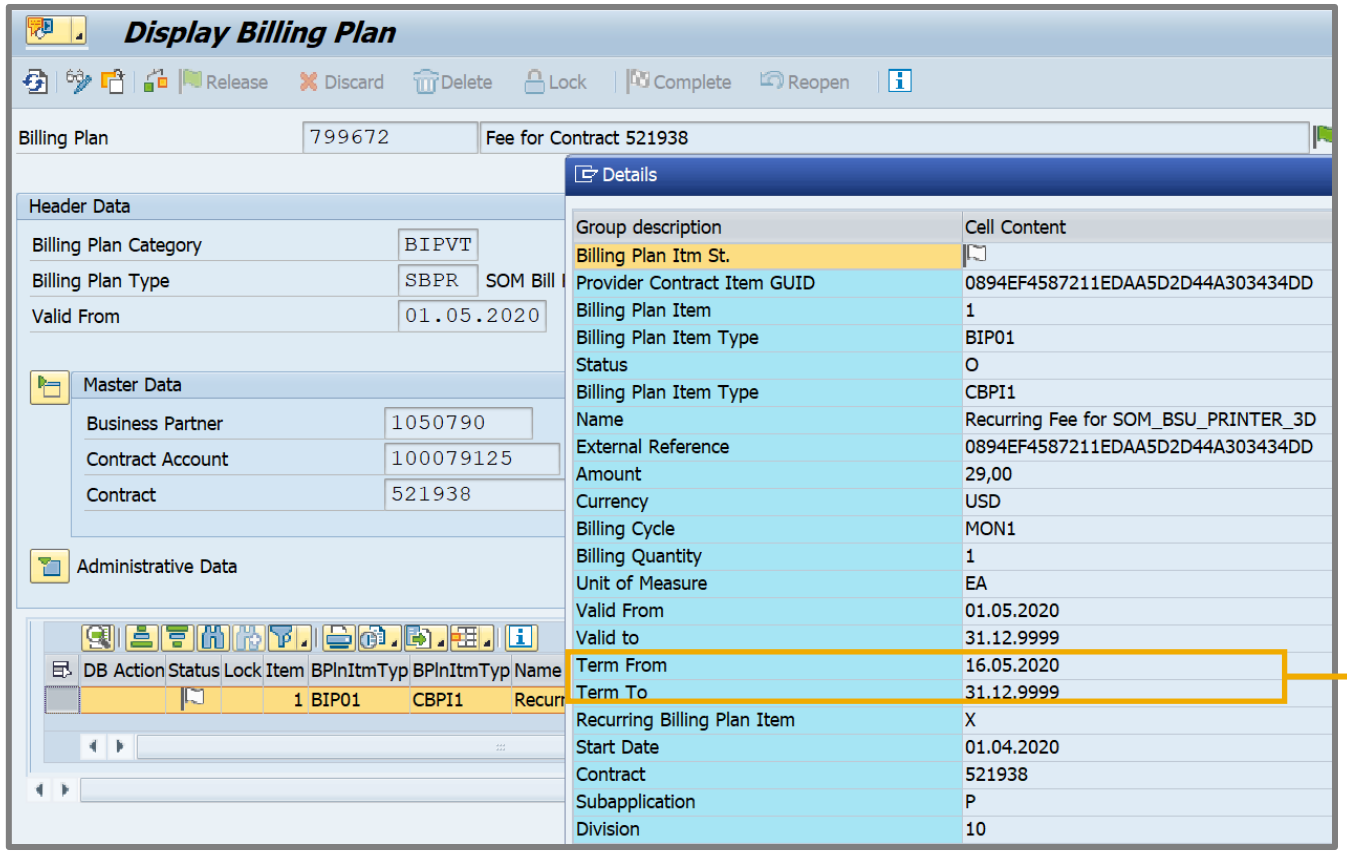
Now we have scenarios where the recurring fees are applicable only once the relevant equipment is activated.

Additionally, it could also be ‘once the equipment is activated’ or ‘X number of days after Contract Start’ whichever is earlier.

To support such scenarios, we now have a new Date Type available (Start of Billing CONTBILS).

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Billing Start Date based on Equipment Activation



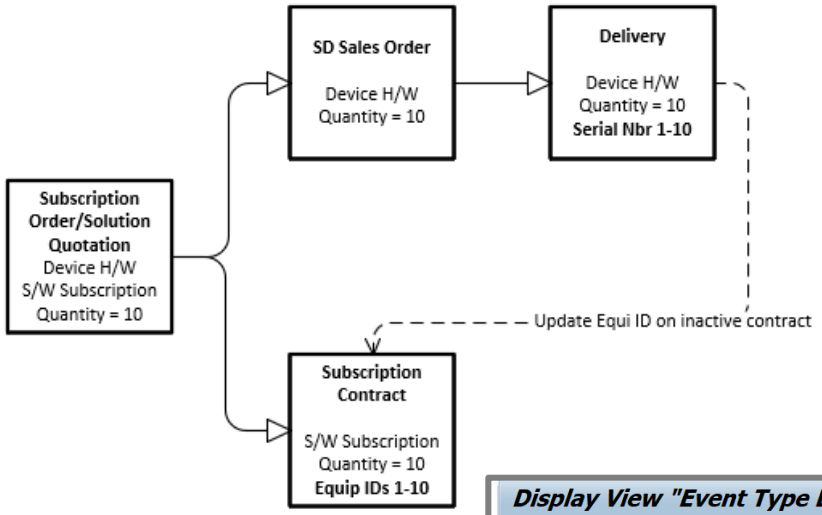
The ‘Start of Billing’ date of the subscription contract is mapped to the ‘Term From’ date of the billing plan item.

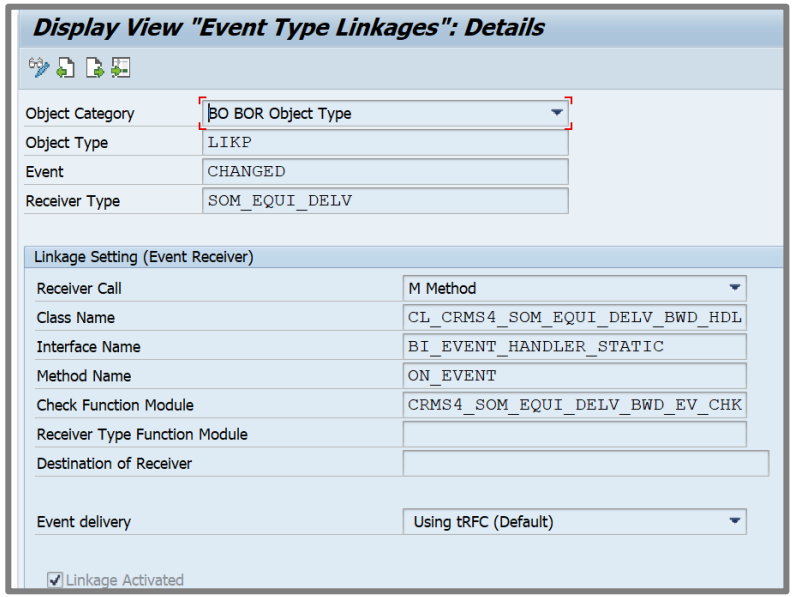
The amount in the BIT that is generated will be prorated based on the ‘Term From’.

Note: in 2020 release there will be a gap with respect to RAR.

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E.g. Equipment IDs maintenance on Contract upon Delivery





When serial numbers are assigned in a delivery and the document is saved, then the event CREATED/CHANGED for the object LIKP is raised

If equipment is created and activated when serial numbers for a product are created, then during delivery it is possible to identify the relevant equipment IDs.

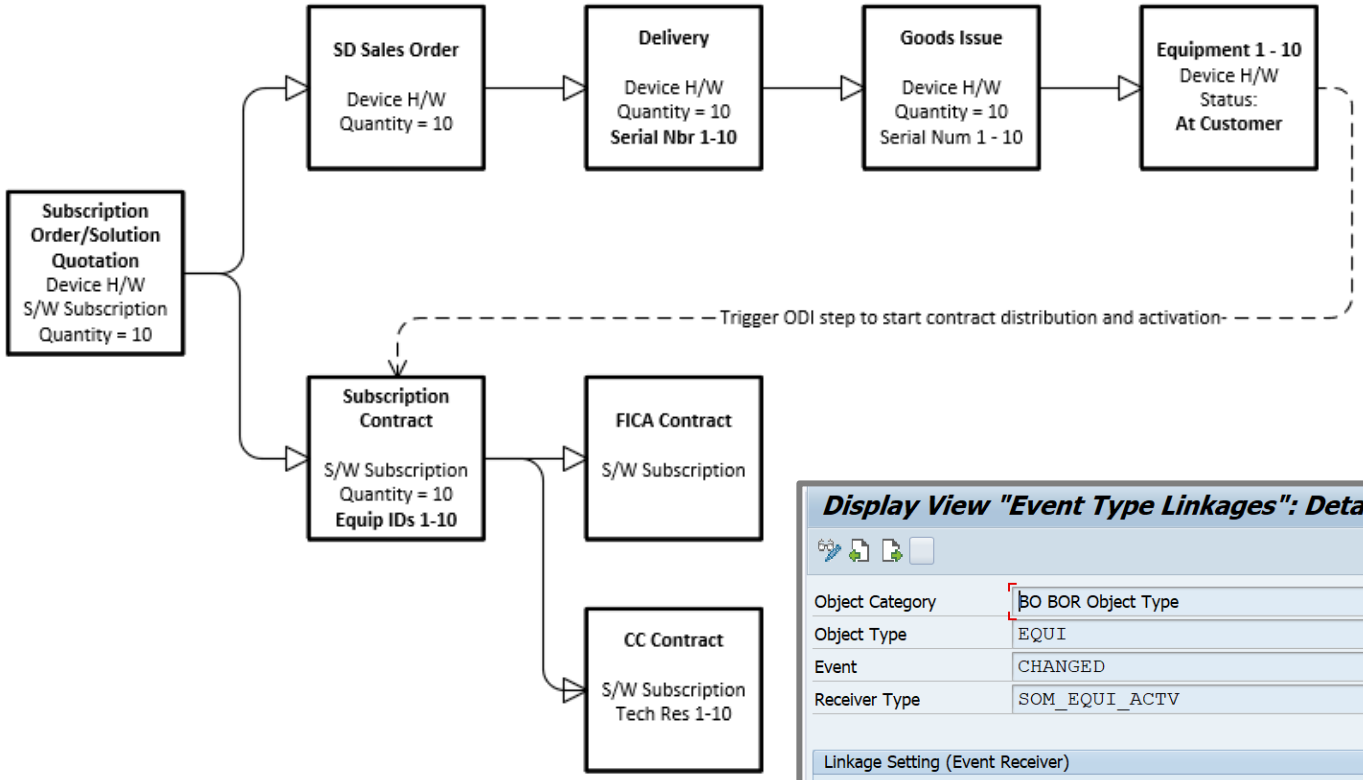
By tracking back from the delivery, it should be possible to identify the relevant Sales item in subscription order / solution quotation.

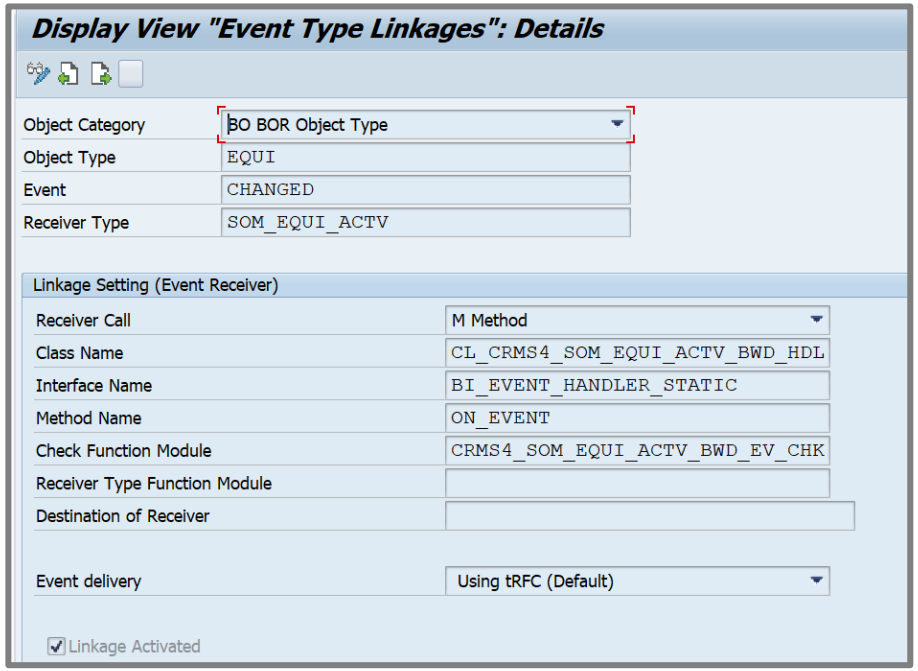
If the products are structured in such a way that the Sales item is a sub-item of a subscription item or vice versa, then that Subscription item is the one on which maintenance of equipment is required.

By registering an event receiver and leveraging the still open for change subscription order or the new ‘Maintain Equipment IDs’ change process, equipment IDs can be maintained on the relevant subscription contract.

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E.g. Equipment activation trigger to start Contract Activation





When equipment status is changed, then the event CHANGED for the object EQUI is raised.

If this equipment, which is changed, is maintained in the object list of a subscription contract, then potentially depending on the status different processes can be triggered on the subscription contract.

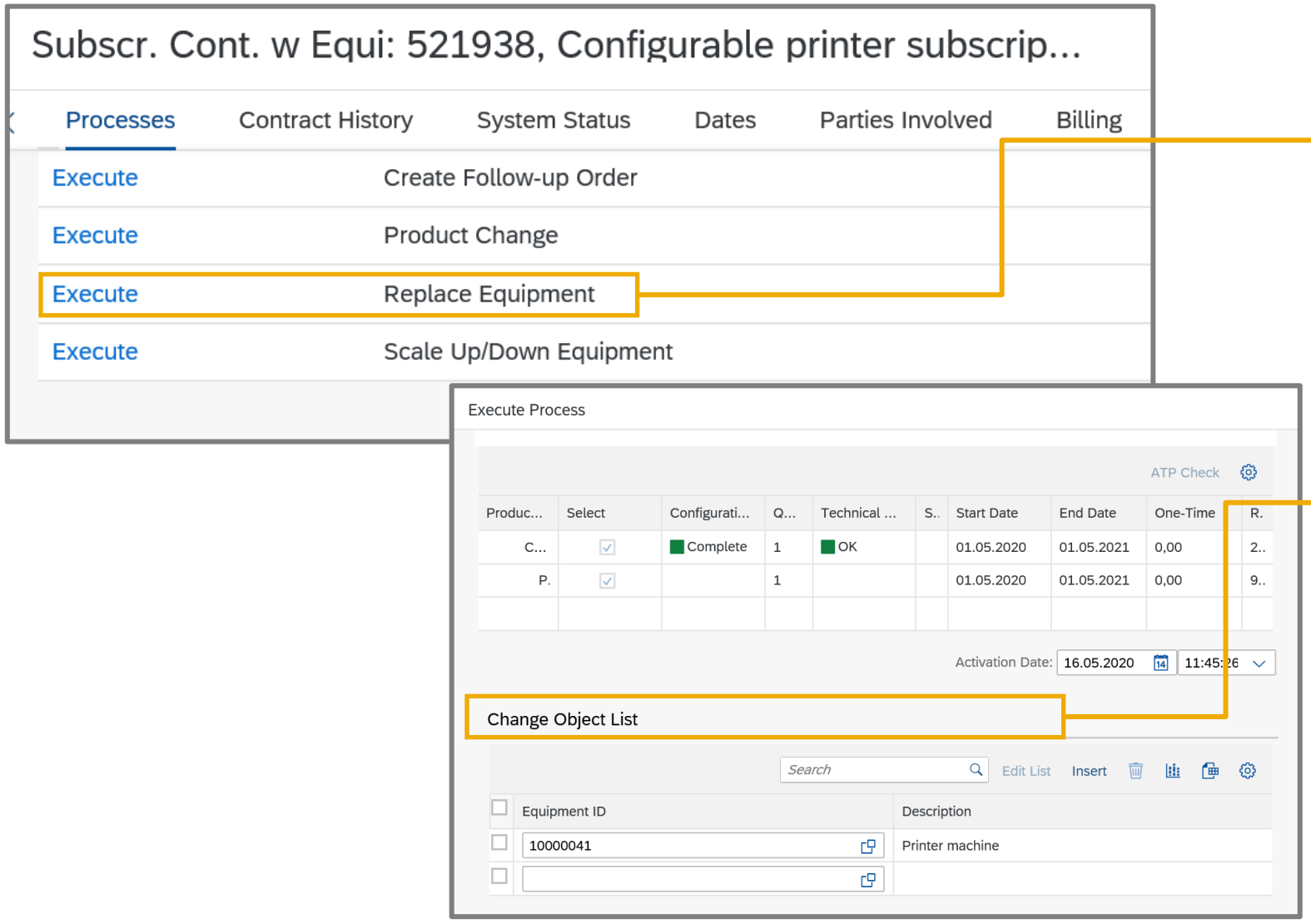
In this example, the status ‘At Customer’ is being checked for and if it is set, an attempt is made to trigger the ODI step of the relevant subscription contract to start the distribution and activation process.

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Device as a Service
Change Processes



Change Processes – Replace Equipment

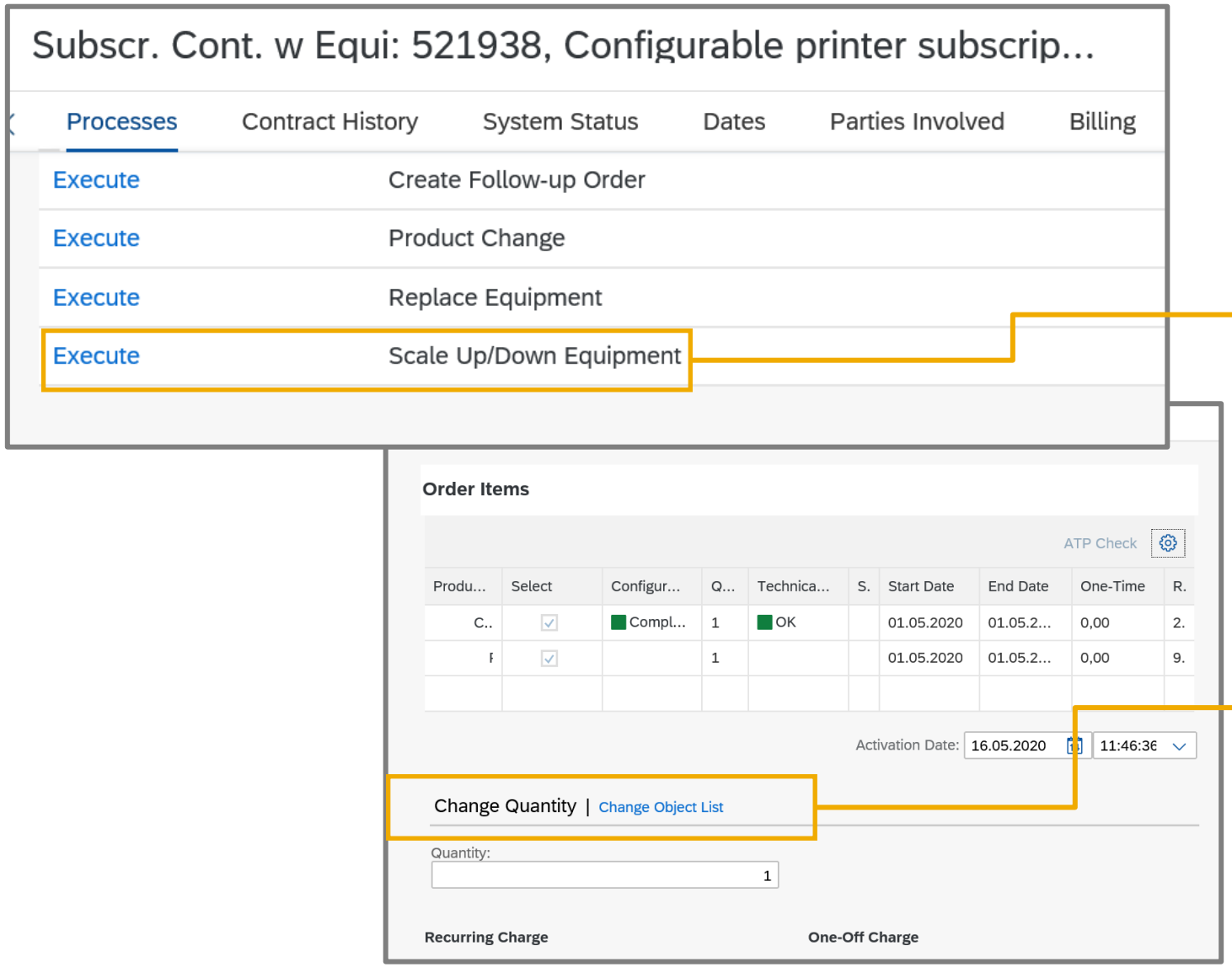


This change process can be used to change the equipment ID(s) on an active subscription contract.

The Object List is shown in the Change Process popup where the list of equipment ID(s) can be maintained.

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Change Processes – Scale Up/Down Equipment



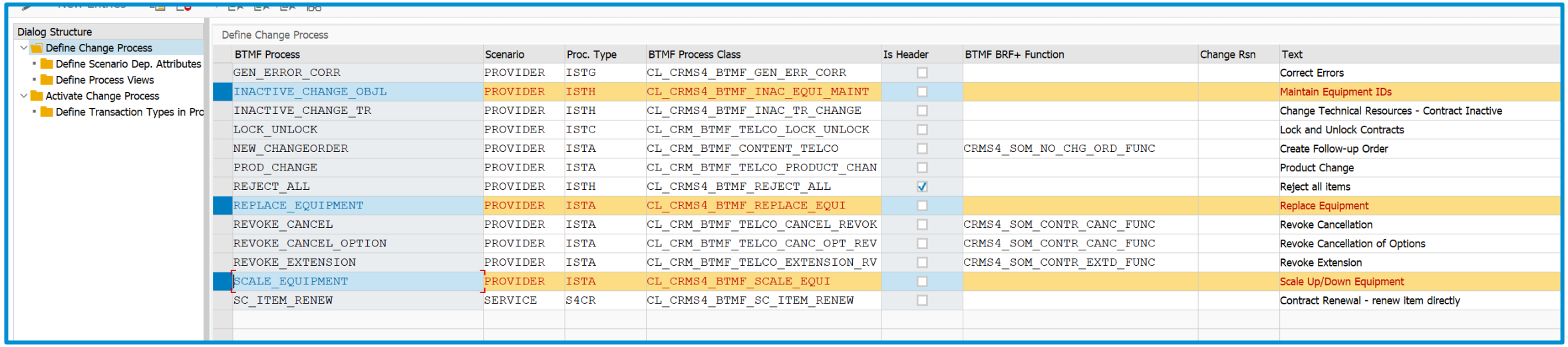
This change process can be used to scale the quantity of the equipment up or down on an active subscription contract.

The Quantity and Object List views are shown in the Change Process popup, where the new Quantity and list of equipment ID(s) can be maintained.

In the case of Scale Up, if the product bundle contains a relevant sales product then in the change order the sales product will be added with the delta quantity.

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Customizing Change Processes



BTMF Process	Scenario	Proc. Type	BTMF Process Class	Is Header	BTMF BRF+ Function	Change Rsn	Text
GEN_ERROR_CORR	PROVIDER	ISTG	CL_CRMS4_BTMF_GEN_ERR_CORR	☐			Correct Errors
INACTIVE_CHANGE_OBJL	PROVIDER	ISTH	CL_CRMS4_BTMF_INAC_EQUI_MAINT	☐			Maintain Equipment IDs
INACTIVE_CHANGE_TR	PROVIDER	ISTH	CL_CRMS4_BTMF_INAC_TR_CHANGE	☐			Change Technical Resources - Contract Inactive
LOCK_UNLOCK	PROVIDER	ISTC	CL_CRM_BTMF_TELCO_LOCK_UNLOCK	☐			Lock and Unlock Contracts
NEW_CHANGEORDER	PROVIDER	ISTA	CL_CRM_BTMF_CONTENT_TELCO	☐	CRMS4_SOM_NO_CHG_ORD_FUNC		Create Follow-up Order
PROD_CHANGE	PROVIDER	ISTA	CL_CRM_BTMF_TELCO_PRODUCT_CHAN	☐			Product Change
REJECT_ALL	PROVIDER	ISTH	CL_CRMS4_BTMF_REJECT_ALL	☑			Reject all items
REPLACE_EQUIPMENT	PROVIDER	ISTA	CL_CRMS4_BTMF_REPLACE_EQUI	☐			Replace Equipment
REVOKE_CANCEL	PROVIDER	ISTA	CL_CRM_BTMF_TELCO_CANCEL_REVOK	☐	CRMS4_SOM_CONTR_CANC_FUNC		Revoke Cancellation
REVOKE_CANCEL_OPTION	PROVIDER	ISTA	CL_CRM_BTMF_TELCO_CANC_OPT_REV	☐	CRMS4_SOM_CONTR_CANC_FUNC		Revoke Cancellation of Options
REVOKE_EXTENSION	PROVIDER	ISTA	CL_CRM_BTMF_TELCO_EXTENSION_RV	☐	CRMS4_SOM_CONTR_EXTD_FUNC		Revoke Extension
SCALE_EQUIPMENT	PROVIDER	ISTA	CL_CRMS4_BTMF_SCALE_EQUI	☐			Scale Up/Down Equipment
SC_ITEM_RENEW	SERVICE	S4CR	CL_CRMS4_BTMF_SC_ITEM_RENEW	☐			Contract Renewal - renew item directly

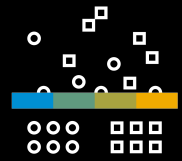
SPRO: Service -> Transactions -> Settings for Subscription Transactions -> Contract Changes -> Settings for the Maintenance Framework (BTMF)

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Miscellaneous








BAdIs

Enhancement Spot:

ES_CRMS4_SOM_EQUI_INTEG

BAdI Definition:

CRMS4_SOM_BADI_EQUI_INTEG

Methods:

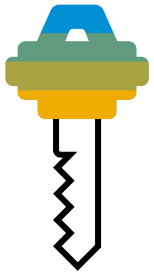
EQUI_STAT_ALLOWS_SUBR_CTR_ACTV	Does Equipment status allow for Subscr. Contract activation?
DETERM_SUBR_CTR_BILLING_START	Determine the Billing Start Date for the Subscr. Contract
DETERM_SUBR_ITEM_BY_SLS_ITEM	Determine the Subcr. item on which Equi is to be maintained
DETERM_TR_IDS_BY_EQUI_IDS	Determine Tech Res IDs based on Equipment Ids

Customizing:

SPRO: Service -> Transactions -> Integration -> Integration with Equipment -> BAdI: Integrate Equipment

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Important Hints



SAP Help

- https://help.sap.com/viewer/ Product SAP S/4HANA
  Enterprise Business Applications -> Service -> Subscription Order Management
- https://help.sap.com/BRIM
  BRIM Scenarios and Guides -> Integration Guide, Migration Guide, Product Page

Application Component

- CRM-S4-SOM

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